

WeatherWise Supply Chain Agent

An AI-powered multi-agent system that detects weather-driven shipment risks, retrieves supplier escalation SOPs, and triggers automated notifications — all from a single natural-language request.

Real-Time Weather Intelligence

Live forecast correlation with in-transit cold-chain shipments for proactive risk detection.

Multi-Agent Orchestration

Five specialized agents coordinate analysis, SOP retrieval, and escalation in a single workflow.

SOP-Aligned Compliance

RAG-powered retrieval ensures every escalation follows the correct supplier playbook.

Automated Escalation

Email reports and SMS alerts triggered automatically with the right contacts at the right tier.

Operational Dashboard

Real-time shipment tracking, carrier status, inventory levels, and supplier overview at a glance.

Natural Language Exploration

Ask Genie enables SQL-powered data exploration through plain English questions.

THE OPPORTUNITY

MedTech Supply Chain Is a Data Problem

Temperature-sensitive shipments face constant weather-driven risk. Today, operations teams manually monitor forecasts, cross-reference shipment data, look up supplier SOPs, and coordinate escalations across email and SMS — a process that takes hours and leaves gaps.

Weather Risk Detection

Correlate live forecasts with shipment routes and temperature thresholds automatically.

Shipment Intelligence

Query in-transit shipments, carriers, and temperature ranges from governed data.

SOP Retrieval

Vector search across supplier escalation playbooks to find the right contacts and procedures.

Automated Notifications

Send detailed escalation emails and concise SMS alerts without manual drafting.

Operational Visibility

Dashboard with shipment status, inventory by site, and supplier health at a glance.

Governance & Audit

Unity Catalog governs all data, tools, and agent traces for full compliance traceability.

APPLICATIONS**Agent Chat**

Conversational interface for weather risk assessment, SOP lookup, and automated escalation.

Supply Chain Dashboard

Real-time shipment tracking, inventory overview, and supplier status in one view.

Ask Genie

Natural language SQL exploration for ad-hoc supply chain questions.

APPLICATION 1 — AGENT CHAT

WeatherWise Escalation Agent

A conversational multi-agent system that orchestrates weather analysis, shipment risk scoring, SOP retrieval, and escalation notifications from a single prompt.

THE CHALLENGE

- Weather-related shipment risks detected too late — hours of manual triage instead of instant alerts
- No automated correlation between live forecasts and in-transit cold-chain shipments
- Escalation workflows are manual — emails, SMS, and supplier SOP lookups handled by separate teams

KEY CAPABILITIES

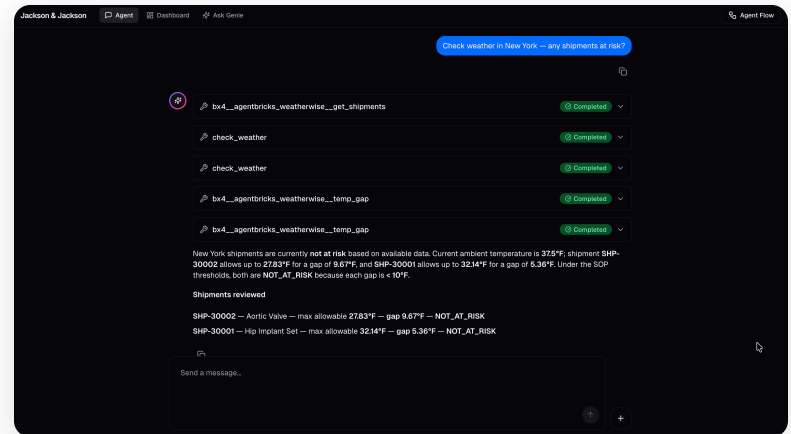
- Live weather-to-shipment risk correlation
- Temperature gap classification and risk scoring
- Supplier SOP retrieval via vector search
- Automated email and SMS escalation
- Multi-agent crew orchestration

DATABRICKS PLATFORM

Mosaic AI Agents LangGraph Unity Catalog

Vector Search MLflow Model Serving

Databricks Apps



AI-POWERED INSIGHT

Five specialized agents — Meteorologist, SQL Analyst, Supplier Researcher, Email Copywriter, and SMS Texter — coordinate in a single natural-language request to detect risk, retrieve SOPs, and trigger escalation in under 30 seconds.

< 30s

End-to-end risk detection and escalation from a single natural-language query

5 Agents

Coordinated crew: weather, data, SOP, email, and SMS agents

100%

Audit-ready traceability with SOP-aligned escalation workflows

APPLICATION 1 — ESCALATION IN ACTION

Automated Email & SMS Alerts

Once risk is assessed, the agent automatically drafts and sends a detailed escalation email and a concise SMS summary — no manual intervention required.

HOW IT WORKS

- Agent assesses shipment risk using weather data and temperature thresholds
- Retrieves the correct supplier SOP and escalation tier contacts
- Drafts a structured email report with cause, data, and recommended actions
- Sends a concise SMS alert to operations teams for immediate awareness
- All actions traced end-to-end via MLflow for audit compliance

ESCALATION DECISION POLICY

- **AT_RISK** — temperature gap $\geq 20^{\circ}\text{F}$ above allowable threshold
- **BORDERLINE** — temperature gap $10\text{--}19^{\circ}\text{F}$ above threshold
- **NOT_AT_RISK** — temperature gap $< 10^{\circ}\text{F}$

INTEGRATIONS

Mailgun Email API

Twilio SMS

Weather API

Unity Catalog Functions

2 Channels

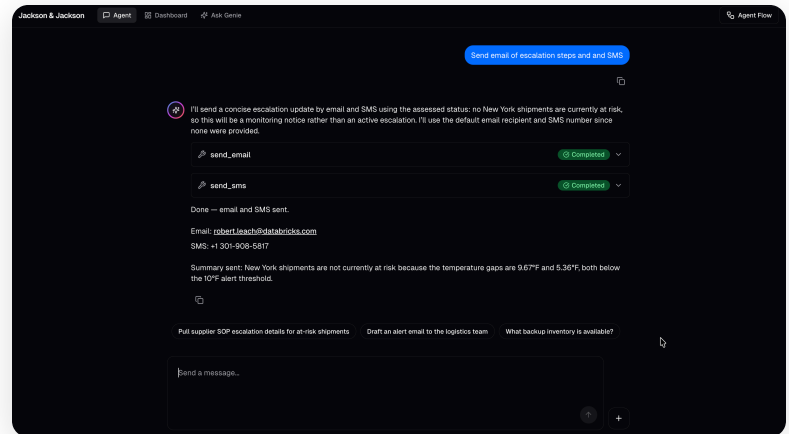
Email for detailed reports, SMS for immediate awareness

Tier-Based

Escalation contacts matched to severity per supplier SOP

Traceable

Every notification logged with MLflow tracing for compliance audit



AUTOMATED ESCALATION

The agent sends a detailed HTML email to the escalation contact and a concise SMS to the operations team — both with risk data, supplier SOP references, and recommended next steps. Default recipients are configured, but can be overridden in the request.

APPLICATION 2 — DASHBOARD

Supply Chain Dashboard

A real-time operational view of shipment status, inventory levels, and supplier health — powered by Unity Catalog governed data and SQL Warehouse.

THE CHALLENGE

- Shipment status scattered across carrier portals with no unified view
- Inventory and supplier data siloed — difficult to assess backup options quickly
- No at-a-glance visibility into delayed, in-transit, or at-risk shipments

KEY CAPABILITIES

- Real-time shipment tracking with status indicators
- Temperature range monitoring per shipment
- Inventory by site with on-hand and lot counts
- Supplier overview with quick-access cards

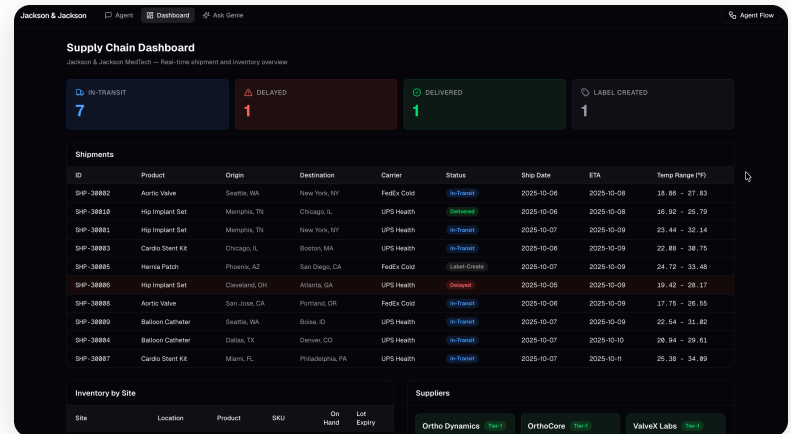
DATABRICKS PLATFORM

SQL Warehouse

Unity Catalog

Databricks Apps

Delta Tables



OPERATIONAL VISIBILITY

Consolidates shipment tracking, carrier status, temperature ranges, inventory levels, and supplier health into a single governed view — eliminating the need to check multiple carrier portals.

1 View

Unified shipment, inventory, and supplier status replacing multiple portals

Real-Time

Live status indicators for in-transit, delayed, and delivered shipments

Governed

All data served through Unity Catalog with role-based access controls

APPLICATION 3 — ASK GENIE

Natural Language SQL Exploration

Ask questions about supply chain data in plain English — Genie translates them to SQL, runs them against governed data, and returns clear answers.

THE CHALLENGE

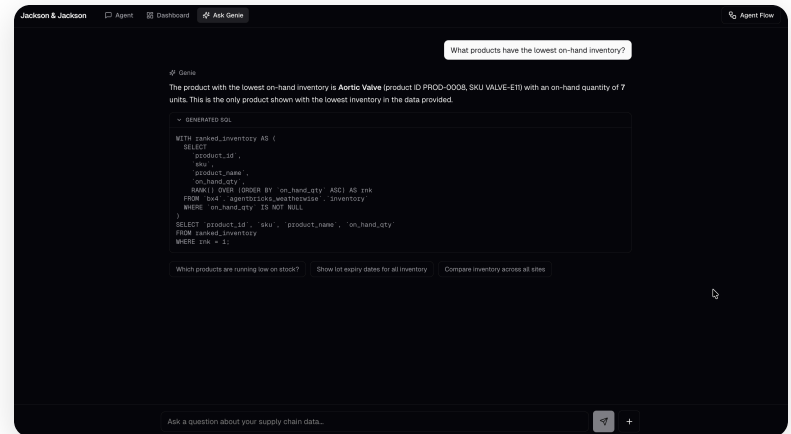
- Ad-hoc supply chain questions require SQL expertise or analyst involvement
- Operations teams wait hours for data answers that should be instant
- Inventory and shipment insights locked behind technical query barriers

KEY CAPABILITIES

- Ask questions in plain English
- Auto-generated SQL with transparent query logic
- Instant answers from governed supply chain data
- Follow-up questions with contextual awareness

DATABRICKS PLATFORM

Genie AI SQL Warehouse Unity Catalog Databricks Apps



AI-POWERED INSIGHT

Genie translates natural language questions into SQL, runs them against governed Unity Catalog data, and returns clear answers — empowering operations teams to self-serve without waiting for analysts.

0 SQL

No SQL expertise required — ask questions in plain English

Instant

Answers in seconds, not hours waiting for analyst support

Transparent

Generated SQL shown for verification and learning

ARCHITECTURE

Platform Reference Architecture

Unity Catalog — Unified Governance

Data, functions, models, and agent tools governed with fine-grained access control and full lineage tracking

DATA SOURCES

Shipments

In-transit status, carrier, ETA, temperature ranges

Suppliers

Contacts, escalation references, SOP documents

Inventory

Warehouse locations, on-hand quantities, lot data

Supplier SOPs

Escalation playbooks indexed for RAG retrieval

Weather API

Live forecast data for shipment destinations

DATABRICKS PLATFORM

Delta Tables — Governed storage for shipments, suppliers, inventory

Vector Search — Supplier SOP index for RAG retrieval

UC Functions — get_shipments, get_supplier_details, temp_gap, get_backup_inventory

Mosaic AI Agents — LangGraph multi-agent orchestration

Model Serving — Serverless agent endpoint with MLflow tracing

MLflow — Agent logging, LLM judge evaluation, UC model registry

SQL Warehouse — Dashboard queries and Genie exploration

APPLICATIONS

Agent Chat

Multi-agent escalation with tool calls, email, and SMS

Dashboard

Real-time shipment, inventory, and supplier view

Ask Genie

Natural language SQL exploration

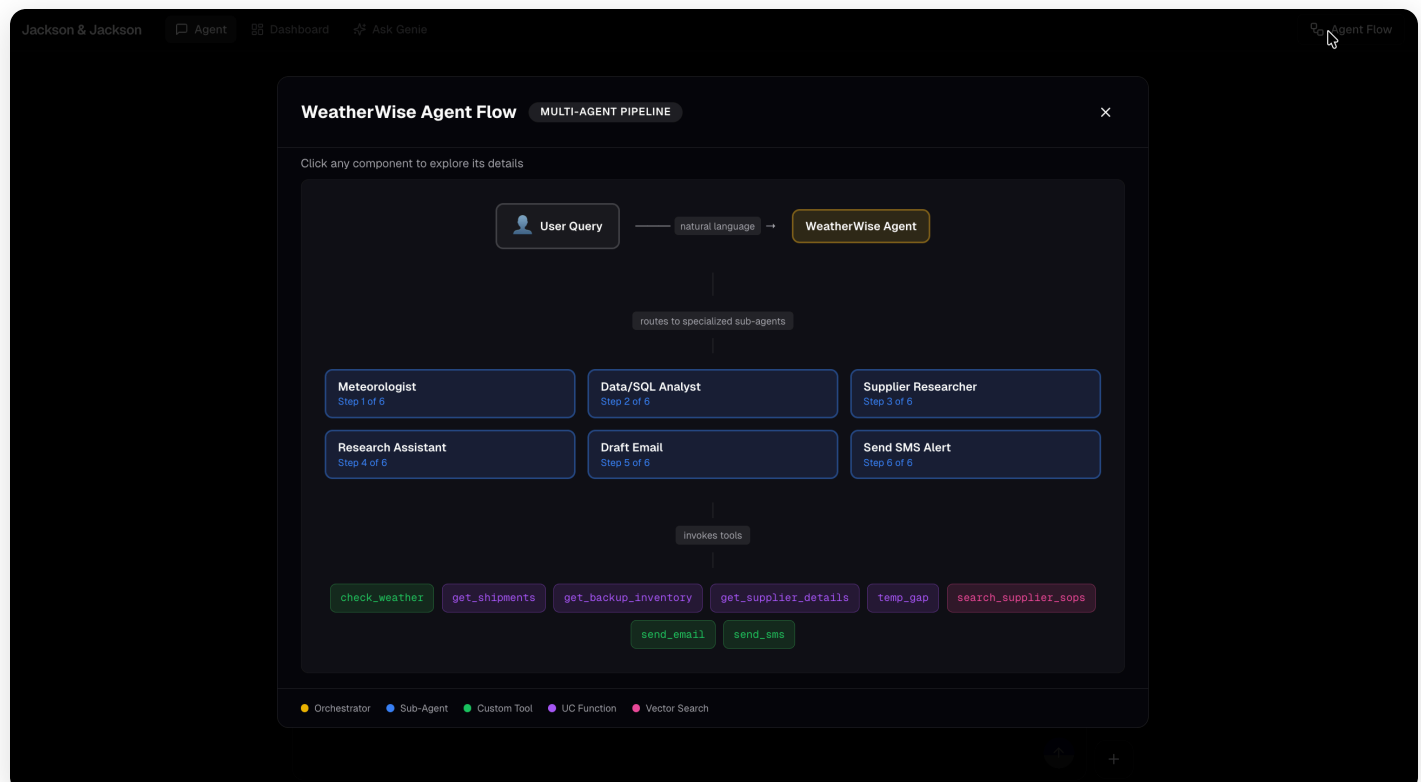
Agent Flow

Visual multi-agent pipeline diagram

Notifications

Mailgun email and Twilio SMS integrations

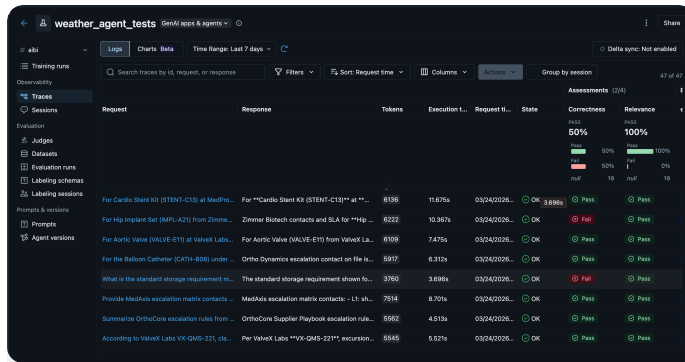
MULTI-AGENT PIPELINE



GOVERNANCE & QUALITY

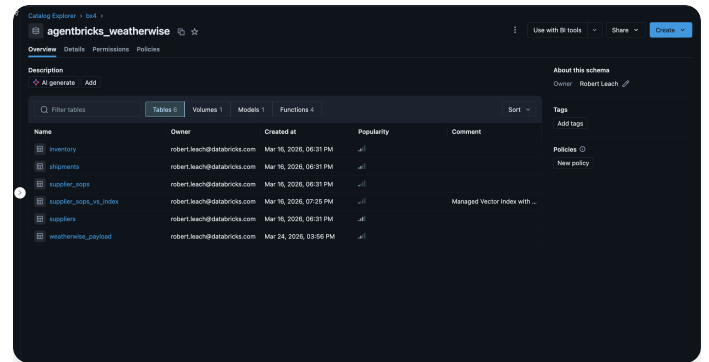
Agent Evaluation & Platform Governance

Every component — data, tools, models, and agent traces — is governed through Unity Catalog. Agent quality is validated through MLflow GenAI evaluation with LLM judges before deployment.



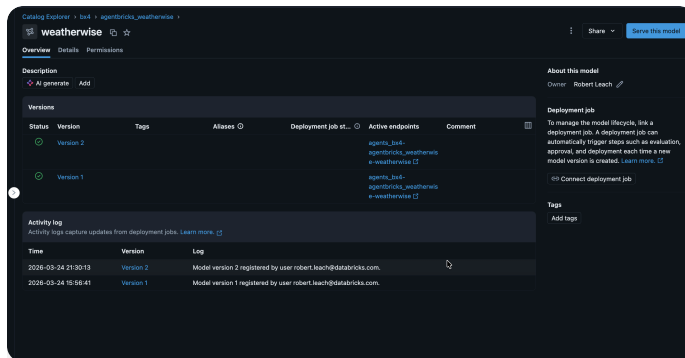
MLflow Agent Evaluation

LLM judges score every response for Correctness, Safety, Relevance, and custom Guidelines before the agent is deployed to production.



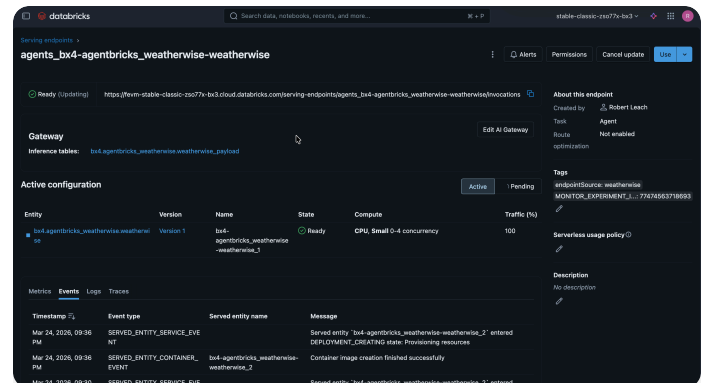
Unity Catalog Schema

Tables, volumes, models, and functions governed in a single catalog with role-based access controls and lineage tracking.



UC Model Registry

Agent versions registered, tracked, and deployed through Unity Catalog with automatic deployment job linking.



Model Serving Endpoint

Serverless deployment with scale-to-zero, inference tables, and real-time monitoring for production agent traffic.

4 Judges

Correctness, Safety, Relevance, and custom Guidelines scored per response

14 Cases

Evaluation dataset covering all supplier SOPs and product-specific scenarios

End-to-End

From data to deployment — governed, evaluated, and traced in one platform